

PROBABILISTIC REASONING

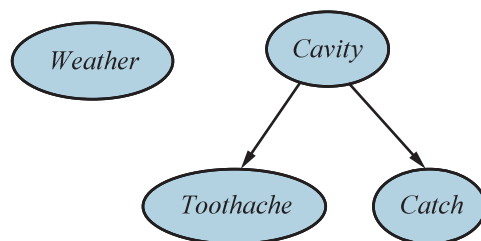


Figure 13.1 simple Bayesian network in which *Weather* is independent of the other three variables and *Toothache* and *Catch* are conditionally independent, given *Cavity*.

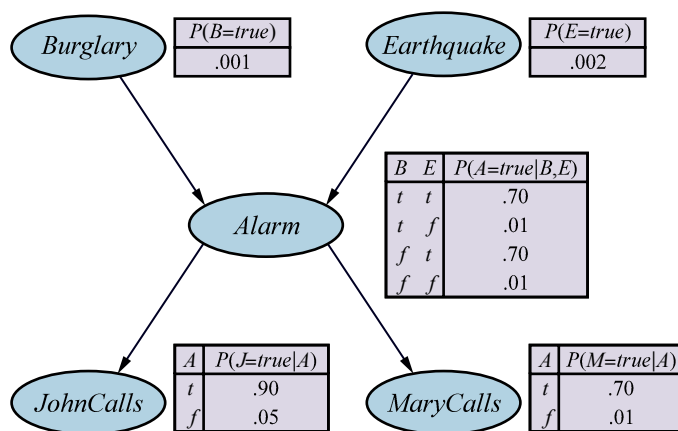


Figure 13.2 typical Bayesian network, showing both the topology and the conditional probability tables (CPTs). In the CPTs, the letters *B*, *E*, *J*, and *M* stand for *Burglary*, *Earthquake*, *Alarm*, *JohnCalls*, and *MaryCalls*, respectively.

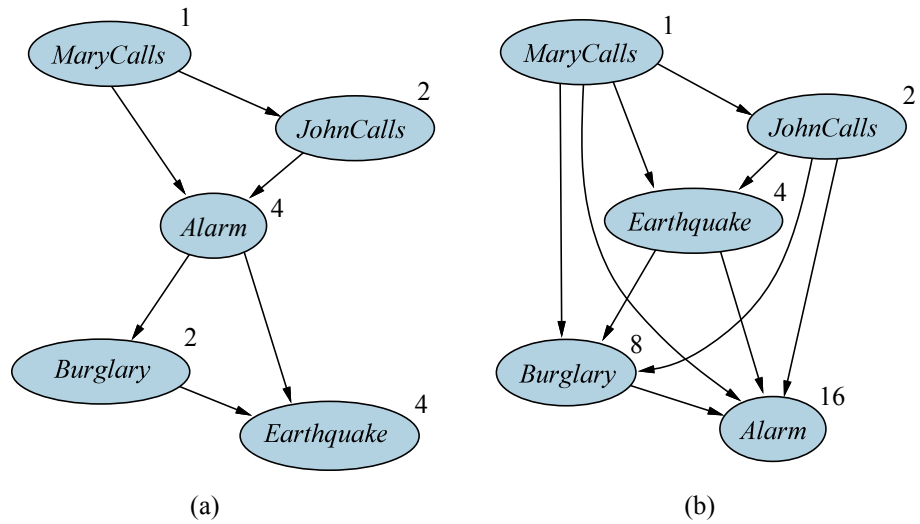


Figure 13.3 Network structure and number of parameters depends on order of introduction. (a) The structure obtained with ordering M, J, B, E . (b) The structure obtained with M, J, E, B . Each node is annotated with the number of parameters required; 13 in all for (a) and 31 for (b). In Figure ??, only 10 parameters were required.

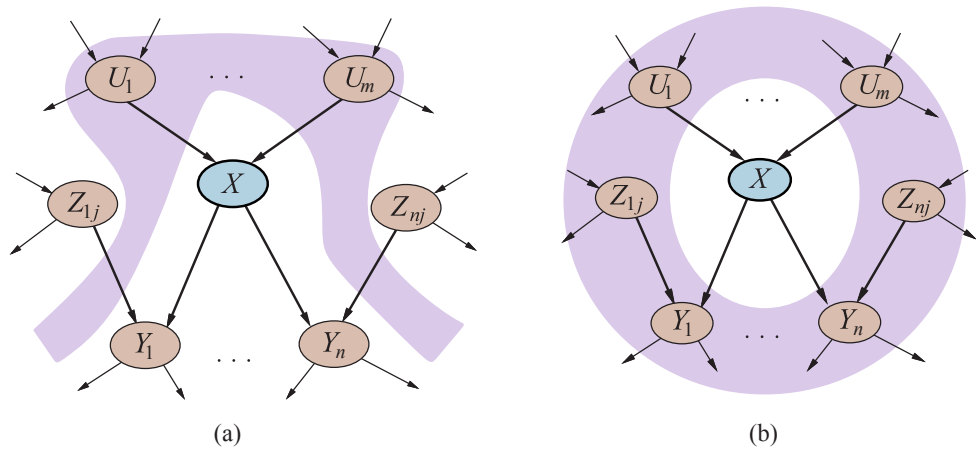


Figure 13.4 (a) node X is conditionally independent of its non-descendants (e.g., the Z_{ij} s) given its parents (the U_i s shown in the gray area). (b) node X is conditionally independent of all other nodes in the network given its Markov blanket (the gray area).

<i>Cold</i>	<i>Flu</i>	<i>Malaria</i>	$P(\text{fever} \mid \cdot)$	$P(\neg \text{fever} \mid \cdot)$
<i>f</i>	<i>f</i>	<i>f</i>	0.0	1.0
<i>f</i>	<i>f</i>	<i>t</i>	0.9	0.1
<i>f</i>	<i>t</i>	<i>f</i>	0.8	0.2
<i>f</i>	<i>t</i>	<i>t</i>	0.98	$0.02 = 0.2 \times 0.1$
<i>t</i>	<i>f</i>	<i>f</i>	0.4	0.6
<i>t</i>	<i>f</i>	<i>t</i>	0.94	$0.06 = 0.6 \times 0.1$
<i>t</i>	<i>t</i>	<i>f</i>	0.88	$0.12 = 0.6 \times 0.2$
<i>t</i>	<i>t</i>	<i>t</i>	0.988	$0.012 = 0.6 \times 0.2 \times 0.1$

Figure 13.5 complete conditional probability table for $\mathbb{P}(\text{Fever} \mid \text{Cold}, \text{Flu}, \text{Malaria})$, assuming a noisy-OR model with the the three q -values shown in bold.

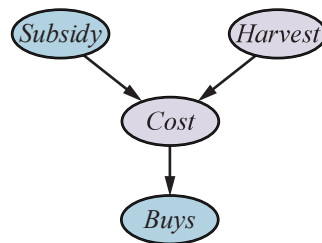


Figure 13.6 simple network with discrete variables (*Subsidy* and *Buys*) and continuous variables (*Harvest* and *Cost*).

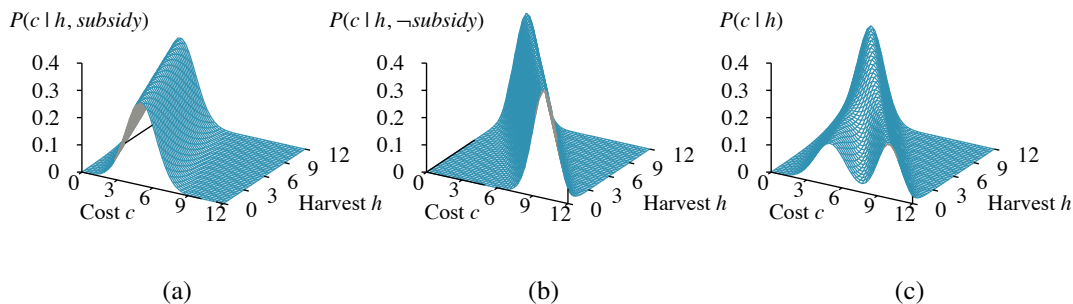


Figure 13.7 The graphs in (a) and (b) show the probability distribution over *Cost* as a function of *Harvest* size, with *Subsidy* true and false, respectively. Graph (c) shows the distribution $P(\text{Cost} \mid \text{Harvest})$, obtained by summing over the two subsidy cases.

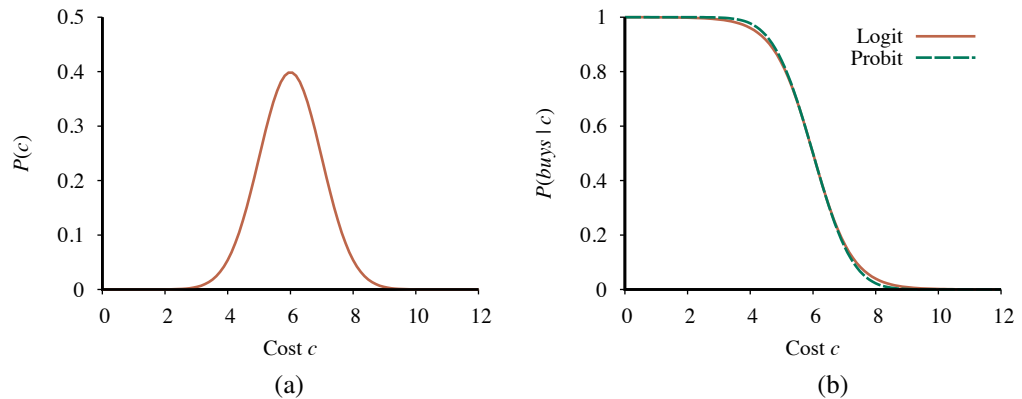


Figure 13.8 (a) normal (Gaussian) distribution for the cost threshold, centered on $\mu = 6.0$ with standard deviation $\sigma = 1.0$. (b) Expit and probit models for the probability of *buys* given *cost*, for the parameters $\mu = 6.0$ and $\sigma = 1.0$.

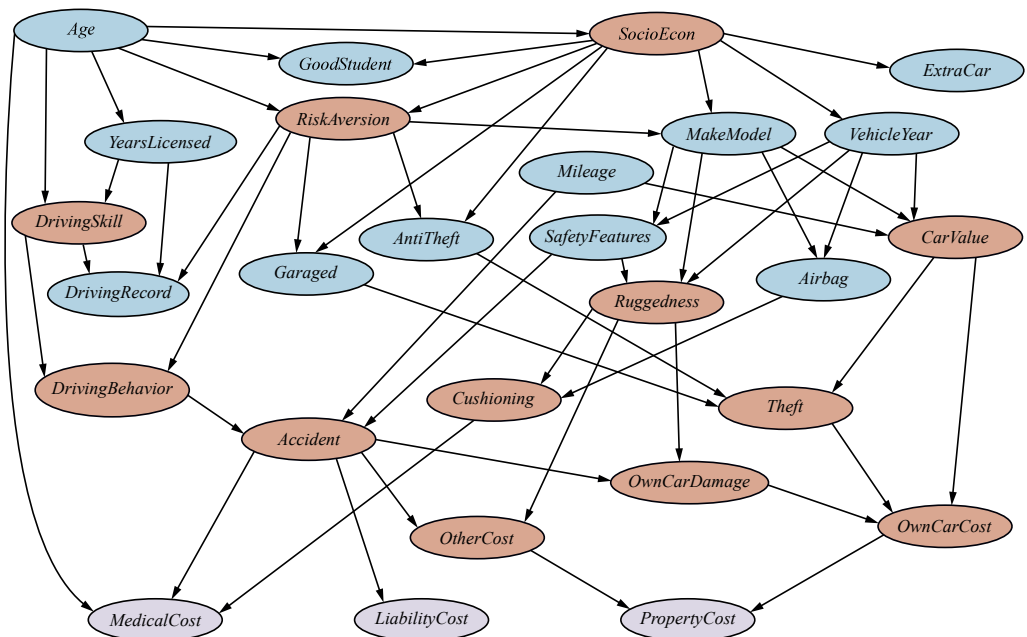


Figure 13.9 Bayesian network for evaluating car insurance applications.

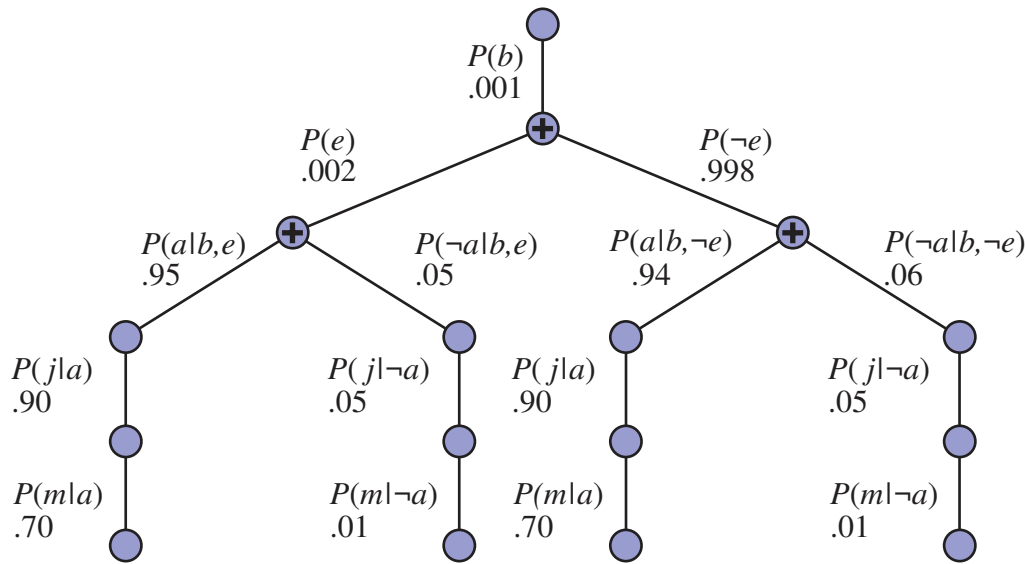


Figure 13.10 The structure of the expression shown in Equation (??). The evaluation proceeds top down, multiplying values along each path and summing at the “+” nodes. Notice the repetition of the paths for j and m .

function ENUMERATION- SK($X, \mathbf{e}, bn$) **returns** a distribution over X

inputs: X , the query variable

$\mathbf{e}$, observed values for variables $\mathbf{E}$

bn , a Bayes net with variables $vars$

$\mathbf{Q}(X) \leftarrow$ a distribution over X , initially empty

for each value x_i of X **do**

$\mathbf{Q}(x_i) \leftarrow$ ENUMERATE- LL($vars, \mathbf{e}_{x_i}$)

where $\mathbf{e}_{x_i}$ is $\mathbf{e}$ extended with $X = x_i$

return NORMALIZE($\mathbf{Q}(X)$)

function ENUMERATE- LL($vars, \mathbf{e}$) **returns** a real number

if EMPTY?($vars$) **then return** 1.0

$V \leftarrow$ FIRST($vars$)

if V is an evidence variable with value v in $\mathbf{e}$

then return $P(v | \text{parents}(V)) \times$ ENUMERATE- LL(REST($vars$), $\mathbf{e}$)

else return $\sum_v P(v | \text{parents}(V)) \times$ ENUMERATE- LL(REST($vars$), $\mathbf{e}_v$)

where $\mathbf{e}_v$ is $\mathbf{e}$ extended with $V = v$

Figure 13.11 The enumeration algorithm for exact inference in Bayes nets.

X	Y	$\mathbf{f}(X, Y)$	Y	Z	$\mathbf{g}(Y, Z)$	X	Y	Z	$\mathbf{h}(X, Y, Z)$
t	t	.3	t	t	.2	t	t	t	$.3 \times .2 = .06$
t	f	.7	t	f	.8	t	t	f	$.3 \times .8 = .24$
f	t	.9	f	t	.6	t	f	t	$.7 \times .6 = .42$
f	f	.1	f	f	.4	t	f	f	$.7 \times .4 = .28$
						f	t	t	$.9 \times .2 = .18$
						f	t	f	$.9 \times .8 = .72$
						f	f	t	$.1 \times .6 = .06$
						f	f	f	$.1 \times .4 = .04$

Figure 13.12 Illustrating pointwise multiplication: $\mathbf{f}(X, Y) \times \mathbf{g}(Y, Z) = \mathbf{h}(X, Y, Z)$.

```
function ELIMINATION- $\text{SK}(X, \mathbf{e}, bn)$  returns a distribution over  $X$ 
  inputs:  $X$ , the query variable
            $\mathbf{e}$ , observed values for variables  $\mathbf{E}$ 
            $bn$ , a Bayesian network with variables  $vars$ 

   $factors \leftarrow []$ 
  for each  $V$  in ORDER( $vars$ ) do
     $factors \leftarrow [\text{MARK-FACTORS}(V, \mathbf{e})] + factors$ 
    if  $V$  is a hidden variable then  $factors \leftarrow \text{SUM-OUT}(V, factors)$ 
  return NORMALIZE(POINTWISE-PRODUCT( $factors$ ))
```

Figure 13.13 The variable elimination algorithm for exact inference in Bayes nets.

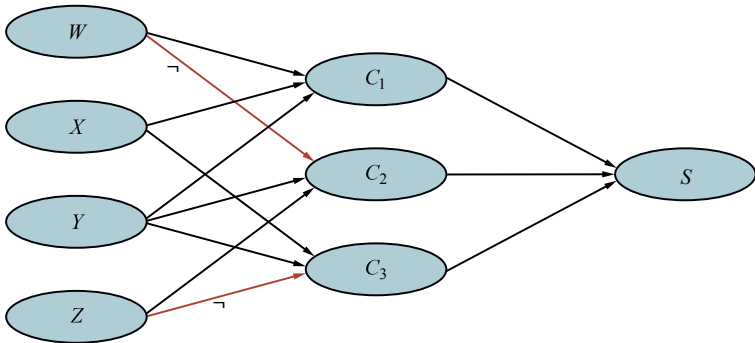


Figure 13.14 Bayes net encoding of the 3-CNF sentence $(W \vee X \vee Y) \wedge (\neg W \vee Y \vee Z) \wedge (X \vee Y \vee \neg Z)$.

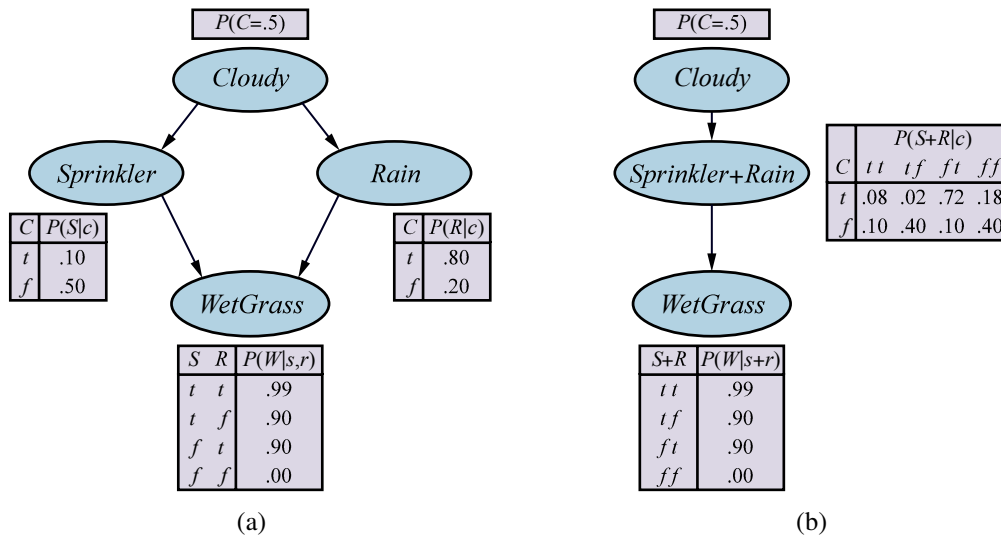


Figure 13.15 (a) multiply connected network describing Mary's daily lawn routine: each morning, she checks the weather; if it's cloudy, she usually doesn't turn on the sprinkler; if the sprinkler is on, or if it rains during the day, the grass will be wet. Thus, *Cloudy* affects *WetGrass* via two different causal pathways. (b) clustered equivalent of the multiply connected network.

function PRIOR-SAMPLE(*bn*) **returns** an event sampled from the prior specified by *bn*
inputs: *bn*, a Bayesian network specifying joint distribution $\mathbf{P}(X_1, \dots, X_n)$
 $\mathbf{x} \leftarrow$ an event with n elements
for each variable X_i **in** $X_1, \dots, X_n$ **do**
 $\mathbf{x}[i] \leftarrow$ a random sample from $\mathbf{P}(X_i \mid \text{parents}(X_i))$
return $\mathbf{x}$

Figure 13.16 sampling algorithm that generates events from a Bayesian network. Each variable is sampled according to the conditional distribution given the values already sampled for the variable's parents.

```

function REJECTION-SAMPLE(X, e, bn, N) returns an estimate of  $P(X | \mathbf{e})$ 
  inputs: X, the query variable
           e, observed values for variables E
           bn, a Bayesian network
           N, the total number of samples to be generated
  local variables: C, a vector of counts for each value of X, initially zero

  for j = 1 to N do
    x ← PRIOR-SAMPLE(bn)
    if x is consistent with e then
      C[j] ← C[j]+1 where  $x_j$  is the value of X in x
  return NORM-LIZE(C)

```

Figure 13.17 The rejection-sampling algorithm for answering queries given evidence in a Bayesian network.

```

function LIKELIHOOD-WEIGHTING(X, e, bn, N) returns an estimate of  $P(X | \mathbf{e})$ 
  inputs: X, the query variable
           e, observed values for variables E
           bn, a Bayesian network specifying joint distribution  $P(X_1, \dots, X_n)$ 
           N, the total number of samples to be generated
  local variables: W, a vector of weighted counts for each value of X, initially zero

  for j = 1 to N do
    x, w ← WEIGHTED-SAMPLE(bn, e)
    W[j] ← W[j] + w where  $x_j$  is the value of X in x
  return NORM-LIZE(W)

function WEIGHTED-SAMPLE(bn, e) returns an event and a weight
  w ← 1; x ← an event with n elements, with values fixed from e
  for i = 1 to n do
    if  $X_i$  is an evidence variable with value  $x_{ij}$  in e
      then w ← w ×  $P(X_i = x_{ij} | \text{parents}(X_i))$ 
      else x[i] ← a random sample from  $P(X_i | \text{parents}(X_i))$ 
  return x, w

```

Figure 13.18 The likelihood-weighting algorithm for inference in Bayesian networks. In WEIGHTED-SAMPLE, each nonevidence variable is sampled according to the conditional distribution given the values already sampled for the variable's parents, while a weight is accumulated based on the likelihood for each evidence variable.

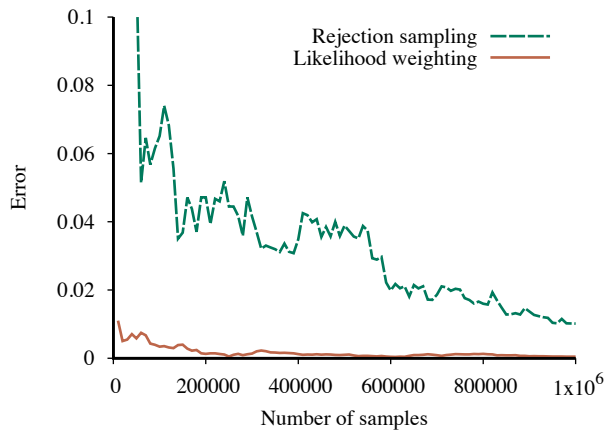


Figure 13.19 Performance of rejection sampling and likelihood weighting on the insurance network. The x-axis shows the number of samples generated and the y-axis shows the maximum absolute error in any of the probability values for a query on *PropertyCost*.

```

function GIBBS- SK( $X, \mathbf{e}, bn, N$ ) returns an estimate of  $\mathbf{P}(X \mid \mathbf{e})$ 
  local variables:  $\mathbf{C}$ , a vector of counts for each value of  $X$ , initially zero
                     $\mathbf{Z}$ , the nonevidence variables in  $bn$ 
                     $\mathbf{x}$ , the current state of the network, initialized from  $\mathbf{e}$ 

  initialize  $\mathbf{x}$  with random values for the variables in  $\mathbf{Z}$ 
  for  $k = 1$  to  $N$  do
    choose any variable  $Z_i$  from  $\mathbf{Z}$  according to any distribution  $\rho(i)$ 
    set the value of  $Z_i$  in  $\mathbf{x}$  by sampling from  $\mathbf{P}(Z_i \mid mb(Z_i))$ 
     $\mathbf{C}[j] \leftarrow \mathbf{C}[j] + 1$  where  $x_j$  is the value of  $X$  in  $\mathbf{x}$ 
  return NORM_LIZE( $\mathbf{C}$ )

```

Figure 13.20 The Gibbs sampling algorithm for approximate inference in Bayes nets; this version chooses variables at random, but cycling through the variables but also works.

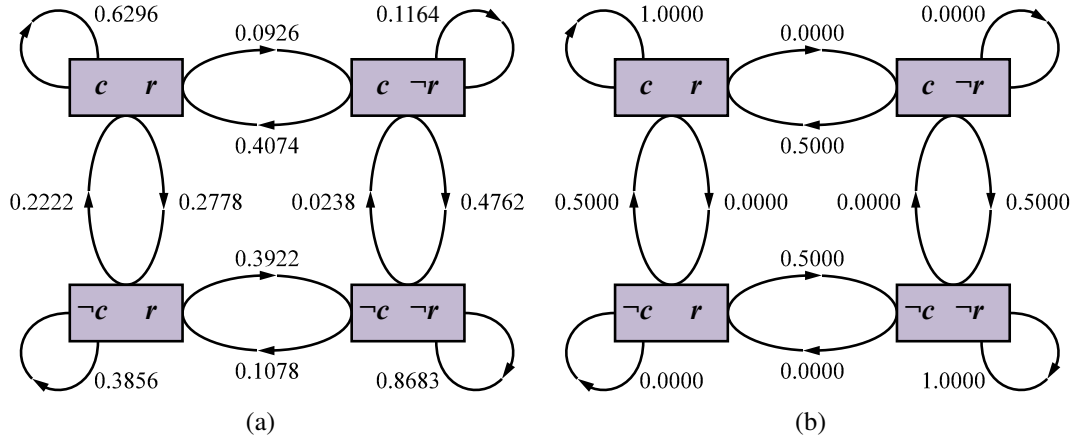


Figure 13.21 (a) The states and transition probabilities of the Markov chain for the query $P(\text{Rain} \mid \text{Sprinkler} = \text{true}, \text{WetGrass} = \text{true})$. Note the self-loops: the state stays the same when *either* variable is chosen and then resamples the same value it already has. (b) The transition probabilities when the CPT for *Rain* constrains it to have the same value as *Cloudy*.

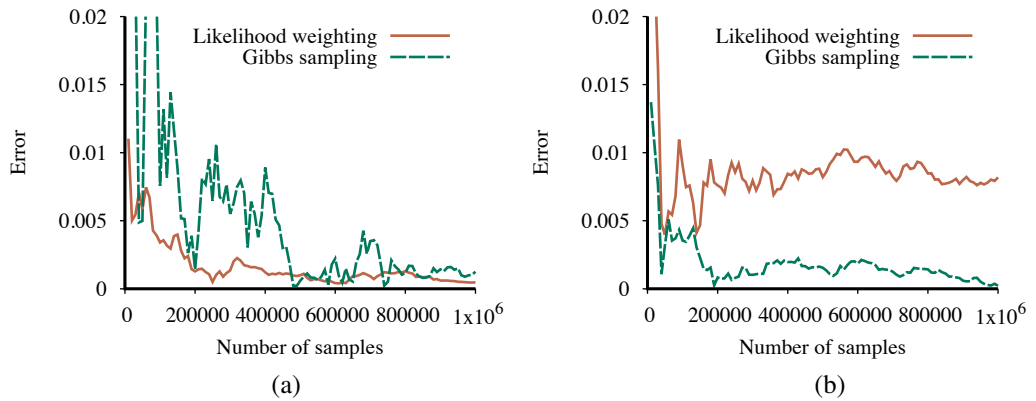


Figure 13.22 Performance of Gibbs sampling compared to likelihood weighting on the car insurance network: (a) for the standard query on *PropertyCost*, and (b) for the case where the output variables are observed and *ge* is the query variable.

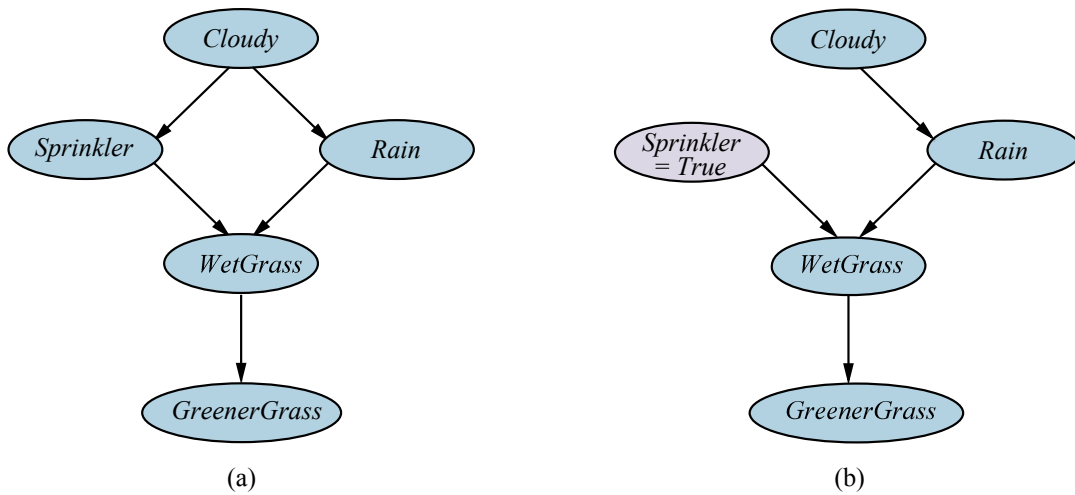


Figure 13.23 (a) causal Bayesian network representing cause–effect relations among five variables. (b) The network after performing the action “turn *Sprinkler* on.”
